Calculus Cheat Sheet

Limits

$$\lim_{h\to 0}\frac{\sin h}{h} = 1.$$

Derivatives

$$\frac{\mathrm{d}x^n}{\mathrm{d}x} = nx^{n-1}.$$

$$\frac{\mathrm{d}|x|}{\mathrm{d}x} = \operatorname{sgn}(x), \quad x \neq 0.$$

$$\frac{\mathrm{d}\ln(x)}{\mathrm{d}x} = \frac{1}{x}.$$

$$\frac{\mathrm{d}a^x}{\mathrm{d}x} = \ln(a)a^x.$$

$$\frac{\mathrm{d}\sin(x)}{\mathrm{d}x} = \cos(x).$$

$$\frac{\mathrm{d}\cos(x)}{\mathrm{d}x} = -\sin(x).$$

$$\frac{\mathrm{d}\tan(x)}{\mathrm{d}x} = \sec^2(x).$$

$$\frac{\mathrm{d}\cot(x)}{\mathrm{d}x} = -\csc^2(x).$$

$$\frac{\mathrm{d}\sec(x)}{\mathrm{d}x} = \tan(x)\sec(x).$$

$$\frac{\mathrm{d}\csc(x)}{\mathrm{d}x} = -\cot(x)\csc(x).$$

$$\frac{\mathrm{d}\arcsin(x)}{\mathrm{d}x} = \frac{1}{\sqrt{1-x^2}}.$$

$$\frac{\mathrm{d}\arccos(x)}{\mathrm{d}x} = -\frac{1}{\sqrt{1-x^2}}.$$

$$\frac{\mathrm{d}\arctan(x)}{\mathrm{d}x} = \frac{1}{1+x^2}.$$

$$\frac{\mathrm{d}\operatorname{arccot}(x)}{\mathrm{d}x} = -\frac{1}{1+x^2}.$$

$$\frac{\mathrm{d}\operatorname{arcsec}(x)}{\mathrm{d}x} = \frac{1}{|x|\sqrt{x^2-1}}.$$

$$\frac{\mathrm{d}\operatorname{arccsc}(x)}{\mathrm{d}x} = -\frac{1}{|x|\sqrt{x^2-1}}.$$

$$\frac{\mathrm{d}\sinh(x)}{\mathrm{d}x} = \cosh(x).$$

$$\frac{\mathrm{d}\cosh(x)}{\mathrm{d}x} = \sinh(x).$$

$$\frac{\mathrm{d}\tanh(x)}{\mathrm{d}x} = \operatorname{sech}^2(x).$$

$$\frac{\mathrm{d}\coth(x)}{\mathrm{d}x} = -\operatorname{csch}^2(x).$$

$$\frac{\mathrm{d}\operatorname{sech}(x)}{\mathrm{d}x} = -\tanh(x)\operatorname{sech}(x).$$

$$\frac{\mathrm{d}\operatorname{csch}(x)}{\mathrm{d}x} = -\coth(x)\operatorname{csch}(x).$$

$$\frac{\mathrm{d}\operatorname{arcsinh}(x)}{\mathrm{d}x} = \frac{1}{\sqrt{x^2+1}}.$$

$$\frac{\mathrm{d}\operatorname{arccosh}(x)}{\mathrm{d}x} = \frac{1}{\sqrt{x^2-1}}.$$

$$\frac{\mathrm{d}\operatorname{arctanh}(x)}{\mathrm{d}x} = \frac{1}{1-x^2}, \quad |x| < 1.$$

$$\frac{\mathrm{d}\operatorname{arccoth}(x)}{\mathrm{d}x} = \frac{1}{1-x^2}, \quad |x| > 1.$$

$$\frac{\mathrm{d}\operatorname{arcsech}(x)}{\mathrm{d}x} = -\frac{1}{x\sqrt{1-x^2}}.$$

$$\frac{\mathrm{d}\operatorname{arccsch}(x)}{\mathrm{d}x} = -\frac{1}{|x|\sqrt{1+x^2}}.$$

Integrals

$$\int x^{-1}\,\mathrm{d}x = \ln|x| + C.$$

$$\int x^n\,\mathrm{d}x = \frac{1}{n+1}x^{n+1} + C, \quad n \neq -1$$

$$\int \ln(x)\,\mathrm{d}x = x\ln(x) - x + C.$$

$$\int \left(\ln(x)\right)^n \mathrm{d}x = x\left(\ln(x)\right)^n - n\int \left(\ln(x)\right)^{n-1}\mathrm{d}x\,, \quad n \in \mathbb{N}.$$

$$\int e^{nx}\,\mathrm{d}x = \frac{1}{n}e^{nx} + C, \quad n \neq 0.$$

$$\int \sin(x)\,\mathrm{d}x = -\cos(x) + C.$$

$$\int \sin^2(x)\,\mathrm{d}x = \frac{x}{2} - \frac{\sin(2x)}{4} + C.$$

$$\int \sin^n(x)\,\mathrm{d}x = -\frac{1}{n}\sin^{n-1}(x)\cos(x) + \frac{n-1}{n}\int \sin^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

$$\int \cos(x)\,\mathrm{d}x = \sin(x) + C.$$

$$\int \cos^2(x)\,\mathrm{d}x = \frac{x}{2} + \frac{\sin(2x)}{4} + C.$$

$$\int \cos^n(x)\,\mathrm{d}x = \frac{1}{n}\cos^{n-1}(x)\sin(x) + \frac{n-1}{n}\int \cos^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

$$\int \tan(x)\,\mathrm{d}x = -\ln|\cos(x)| + C = \ln|\sec(x)| + C.$$

$$\int \tan^2(x)\,\mathrm{d}x = \tan(x) - x + C.$$

$$\int \tan^n(x)\,\mathrm{d}x = \frac{1}{n-1}\tan^{n-1}(x) - \int \tan^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

$$\int \cot(x)\,\mathrm{d}x = \ln|\sin(x)| + C.$$

$$\int \cot^2(x)\,\mathrm{d}x = -\cot(x) - x + C.$$

$$\int \cot^n(x)\,\mathrm{d}x = -\frac{1}{n-1}\cot^{n-1}(x) - \int \cot^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

$$\int \sec(x)\,\mathrm{d}x = \ln|\tan(x) + \sec(x)| + C = \ln\left|\tan\left(\frac{x}{2} + \frac{\pi}{4}\right)\right| + C.$$

$$\int \sec^2(x)\,\mathrm{d}x = \tan(x) + C.$$

$$\int \sec^n(x)\,\mathrm{d}x = \frac{1}{n-1}\tan(x)\sec^{n-2}(x) + \frac{n-2}{n-1}\int \sec^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

$$\int \csc(x)\,\mathrm{d}x = -\ln|\cot(x) + \csc(x)| + C = -\ln\left|\cot\left(\frac{x}{2}\right)\right| + C.$$

$$\int \csc^2(x)\,\mathrm{d}x = -\cot(x) + C.$$

$$\int \csc^n(x)\,\mathrm{d}x = -\frac{1}{n-1}\cot(x)\csc^{n-2}(x) + \frac{n-2}{n-1}\int \csc^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

$$\int \arcsin(x)\,\mathrm{d}x = x\arcsin(x) + \sqrt{1-x^2} + C.$$

$$\int \arccos(x)\,\mathrm{d}x = x\arccos(x) - \sqrt{1-x^2} + C.$$

$$\int \arctan(x)\,\mathrm{d}x = x\arctan(x) - \frac{1}{2}\ln\left(1+x^2\right) + C.$$

$$\int \operatorname{arccot}(x)\,\mathrm{d}x = x\operatorname{arccot}(x) + \frac{1}{2}\ln\left(1+x^2\right) + C.$$

$$\int \operatorname{arcsec}(x)\,\mathrm{d}x = x\operatorname{arcsec}(x) - \operatorname{sgn}(x)\ln\left|x + \sqrt{x^2-1}\right| + C, \quad |x| \geq 1.$$

$$\int \operatorname{arccsc}(x)\,\mathrm{d}x = x\operatorname{arccsc}(x) + \operatorname{sgn}(x)\ln\left|x + \sqrt{x^2-1}\right| + C, \quad |x| \geq 1.$$

$$\int \sinh(x)\,\mathrm{d}x = \cosh(x) + C.$$

$$\int \sinh^2(x)\,\mathrm{d}x = -\frac{x}{2} + \frac{\sinh(2x)}{4} + C.$$

$$\int \sinh^n(x)\,\mathrm{d}x = \frac{1}{n}\sinh^{n-1}(x)\cosh(x) - \frac{n-1}{n}\int \sinh^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

$$\int \cosh(x)\,\mathrm{d}x = \sinh(x) + C.$$

$$\int \cosh^2(x)\,\mathrm{d}x = \frac{x}{2} + \frac{\sinh(2x)}{4} + C.$$

$$\int \cosh^n(x)\,\mathrm{d}x = \frac{1}{n}\cosh^{n-1}(x)\sinh(x) + \frac{n-1}{n}\int \cosh^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

$$\int \tanh(x)\,\mathrm{d}x = \ln\big(\cosh(x)\big) + C.$$

$$\int \tanh^2(x)\,\mathrm{d}x = x - \tanh(x) + C.$$

$$\int \tanh^n(x)\,\mathrm{d}x = \frac{1}{n-1}\tanh^{n-1}(x) - \int \tanh^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

$$\int \coth(x)\,\mathrm{d}x = \ln\big|\sinh(x)\big| + C.$$

$$\int \coth^2(x)\,\mathrm{d}x = x - \coth(x) + C.$$

$$\int \coth^n(x)\,\mathrm{d}x = -\frac{1}{n-1}\coth^{n-1}(x) - \int \coth^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

$$\int \operatorname{sech}(x)\,\mathrm{d}x = 2\arctan\Big(\tanh\Big(\frac{x}{2}\Big)\Big) + C = \arctan\big(\sinh(x)\big) + C = \arcsin\big(\tanh(x)\big) + C.$$

$$\int \operatorname{sech}^2(x)\,\mathrm{d}x = \tanh(x) + C.$$

$$\int \operatorname{sech}^n(x)\,\mathrm{d}x = \frac{1}{n-1}\tanh(x)\operatorname{sech}^{n-1}(x) + \frac{n-2}{n-1}\int \operatorname{sech}^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

$$\int \operatorname{csch}(x)\,\mathrm{d}x = \ln\Big|\tanh\Big(\frac{x}{2}\Big)\Big| + C = \ln\big|\coth(x) - \operatorname{csch}(x)\big| + C.$$

$$\int \operatorname{csch}^2(x)\,\mathrm{d}x = -\coth(x) + C.$$

$$\int \operatorname{csch}^n(x)\,\mathrm{d}x = -\frac{1}{n-1}\coth(x)\operatorname{csch}^{n-1}(x) + \frac{n-2}{n-1}\int \operatorname{csch}^{n-2}(x)\,\mathrm{d}x\,, \quad n \in \mathbb{N}_{>2}.$$

$$\int \operatorname{arcsinh}(x)\,\mathrm{d}x = x\operatorname{arcsinh}(x) - \sqrt{x^2+1} + C.$$

$$\int \operatorname{arccosh}(x)\,\mathrm{d}x = x\operatorname{arccosh}(x) - \sqrt{x^2-1} + C.$$

$$\int \operatorname{arctanh}(x)\,\mathrm{d}x = x\operatorname{arctanh}(x) + \frac{1}{2}\ln\big(1-x^2\big) + C.$$

$$\int \operatorname{arccoth}(x)\,\mathrm{d}x = x\operatorname{arccoth}(x) + \frac{1}{2}\ln\big(x^2-1\big) + C.$$

$$\int \operatorname{arcsech}(x)\,\mathrm{d}x = x\operatorname{arcsech}(x) + \arctan\left(\frac{\sqrt{1-x^2}}{x}\right) + C.$$

$$\int \operatorname{arccsch}(x)\,\mathrm{d}x = x\operatorname{arccsch}(x) + \ln\Big|x + \sqrt{1+x^2}\Big| + C.$$

$$\int \frac{1}{x^2+a^2}\,\mathrm{d}x = \frac{1}{a}\arctan\Big(\frac{x}{a}\Big) + C, \quad a > 0.$$

$$\int \frac{1}{x^2-a^2}\,\mathrm{d}x = \frac{1}{2a}\ln\Big(\frac{x-a}{x+a}\Big) + C = \frac{1}{a}\operatorname{arccoth}\Big(\frac{x}{a}\Big) + C, \quad a > 0 \wedge x^2 - a^2 > 0.$$

$$\int \frac{1}{a^2-x^2}\,\mathrm{d}x = \frac{1}{2a}\ln\Big(\frac{a-x}{a+x}\Big) + C = \frac{1}{a}\operatorname{arctanh}\Big(\frac{x}{a}\Big) + C, \quad a > 0 \wedge a^2 - x^2 > 0.$$

$$\int \frac{1}{ax+b}\,\mathrm{d}x = \frac{1}{a}\ln|ax+b| + C, \quad a > 0.$$

$$\int (ax+b)^n\,\mathrm{d}x = \frac{1}{a(n+1)}(ax+b)^{n+1} + C, \quad a \neq 0 \wedge n \neq -1.$$

$$\int \frac{x}{ax+b}\,\mathrm{d}x = \frac{x}{a} - \frac{b}{a^2}\ln|ax+b| + C, \quad a > 0.$$

$$\int \frac{x}{(ax+b)^2}\,\mathrm{d}x = \frac{1}{a^2}\ln|ax+b| + \frac{b}{a^2(ax+b)} + C, \quad a > 0.$$

$$\int x(ax+b)^n\,\mathrm{d}x = \frac{a(n+1)x-b}{a^2(n+1)(n+2)}(ax+b)^{n+1} + C, \quad a \neq 0 \wedge n \notin \{-1,-2\}.$$

Let $r = \sqrt{a^2+x^2}$.

$$\int r\,\mathrm{d}x = \frac{1}{2}\left(xr + a^2\ln(x+r)\right) + C.$$

$$\int xr\,\mathrm{d}x = \frac{1}{3}r^3 + C.$$

$$\int \frac{1}{r}\,\mathrm{d}x = \operatorname{arcsinh}\left(\frac{x}{|a|}\right) + C = \ln(x+r) + C.$$

$$\int \frac{x}{r}\,\mathrm{d}x = r + C.$$

Let $s = \sqrt{x^2-a^2}, \quad x^2 > a^2$.

$$\int s\,\mathrm{d}x = \frac{1}{2}\left(xs - a^2\ln(x+s)\right) + C.$$

$$\int xs\,\mathrm{d}x = \frac{1}{3}s^3 + C.$$

$$\int \frac{1}{s}\,\mathrm{d}x = \arcsin\left(\frac{x}{|a|}\right) + C.$$

$$\int \frac{x}{s}\,\mathrm{d}x = -s + C.$$

Let $q = \sqrt{a^2-x^2}, \quad a^2 \geq x^2$.

$$\int q\,\mathrm{d}x = \frac{1}{2}\left(q + a^2\arcsin\left(\frac{x}{|a|}\right)\right) + C.$$

$$\int xq\,\mathrm{d}x = \frac{1}{3}q^3 + C.$$

$$\int \frac{1}{q}\,\mathrm{d}x = \arcsin\left(\frac{x}{|a|}\right) + C.$$

$$\int \frac{x}{q}\,\mathrm{d}x = q + C.$$

Let $R = \sqrt{ax+b}, \quad a \neq 0$.

$$\int R\,\mathrm{d}x = \frac{2}{3a}R^3 + C.$$

$$\int x^n R\,\mathrm{d}x = \frac{2}{a(2n+3)}\left(x^n R^3 - bn\int x^{n-1}R\,\mathrm{d}x\right), \quad n \in \mathbb{N}.$$

$$\int \frac{1}{R}\,\mathrm{d}x = \frac{2R}{a} + C.$$

$$\int \frac{x^n}{R}\,\mathrm{d}x = \frac{2}{a}\left(x^n R - n\int x^{n-1}R\,\mathrm{d}x\right) = \frac{2}{a(2n+1)}\left(x^n R - bn\int \frac{x^{n-1}}{R}\,\mathrm{d}x\right), \quad n \in \mathbb{N}.$$

Maclaurin Series and Taylor Series

$$x^r = \sum_{n=0}^{\infty} \binom{r}{n} a^{r-n}(x-a)^n, \quad |x-a| < |a|.$$

$$(1+x)^r = \sum_{n=0}^{\infty} \binom{r}{n} x^n, \quad r \in \mathbb{N}_0 \vee -1 < x < 1 \vee (-1 \leq x \leq 1 \wedge r > 0) \vee (x = 1 \wedge -1 < r < 0).$$

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-x)^n, \quad |x| < 1.$$

$$\frac{1}{1+x} = \frac{1}{1+a} \sum_{n=0}^{\infty} \left(\frac{a-x}{1+a}\right)^n, \quad |x-a| < |1+a|.$$

$$\sqrt{a^2+x^2} = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}(2n)!}{4^n(n!)^2(2n-1)} \frac{x^{2n}}{a^{2n-1}}, \quad |x| < |a|.$$

$$(1-x)^r = \sum_{n=0}^{\infty} \binom{r}{n} (-x)^n, \quad r \in \mathbb{N}_0 \vee -1 < x < 1 \vee (-1 \leq x \leq 1 \wedge r > 0) \vee (x = -1 \wedge -1 < r < 0).$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1.$$

$$\frac{1}{1-x} = \frac{1}{1-a} \sum_{n=0}^{\infty} \left(\frac{x-a}{1-a}\right)^n, \quad |x-a| < |1-a|.$$

$$\sqrt{a^2-x^2} = -\sum_{n=0}^{\infty} \frac{(2n)!}{4^n(n!)^2(2n-1)} \frac{x^{2n}}{a^{2n-1}}, \quad |x| < |a|.$$

$$\ln(x) = \ln(a) + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \left(\frac{x-a}{a}\right)^n, \quad 0 \leq x \leq 2a \wedge a > 0.$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n, \quad -1 < x \leq 1.$$

$$\ln(1+x) = \ln(1+a) + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \left(\frac{x-a}{1+a}\right)^n, \quad -1 < x \leq 1+2a \wedge a > -1.$$

$$\ln(1-x) = -\sum_{n=1}^{\infty} \frac{1}{n} x^n, \quad -1 \leq x < 1.$$

$$\ln(1-x) = \ln(1-a) - \sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{x-a}{1-a}\right)^n, \quad 2a-1 \leq x < 1 \wedge a > 1.$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

$$e^x = e^a \sum_{n=0}^{\infty} \frac{(x-a)^n}{n!}.$$

$$b^x = \sum_{n=0}^{\infty} \frac{(\ln b)^n}{n!} x^n, \quad b > 0.$$

$$b^x = b^a \sum_{n=0}^{\infty} \frac{(\ln b)^n}{n!}(x-a)^n, \quad b > 0.$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}.$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{\sin\left(a + \frac{n\pi}{2}\right)}{n!}(x-a)^n.$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}.$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{\cos\left(a + \frac{n\pi}{2}\right)}{n!}(x-a)^n.$$

$$\arcsin(x) = \sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1}, \quad |x| \leq 1.$$

$$\arccos(x) = \frac{\pi}{2} - \sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2 (2n+1)} x^{2n+1}, \quad |x| \leq 1.$$

$$\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}, \quad |x| \leq 1.$$

$$\operatorname{arccot}(x) = \frac{\pi}{2} - \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}, \quad |x| \leq 1.$$

$$\sinh(x) = \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n+1}.$$

$$\sinh(x) = \sum_{n=0}^{\infty} \frac{((n+1) \mod 2)\sinh a + (n \mod 2)\cosh a}{n!}(x-a)^n.$$

$$\cosh(x) = \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n}.$$

$$\cosh(x) = \sum_{n=0}^{\infty} \frac{((n+1) \mod 2)\cosh a + (n \mod 2)\sinh a}{n!}(x-a)^n.$$

Multivariate Calculus

$$\int_C \nabla \mathbf{F} \, \mathrm{d}\mathbf{r} = \mathbf{F}(\mathbf{r}(b)) - \mathbf{F}(\mathbf{r}(a)), \quad C: \mathbf{r}(t), \quad t \in [a, b].$$

$$\int_V \nabla \cdot \mathbf{F} \, \mathrm{d}V = \oint_S \mathbf{F} \cdot \mathrm{d}\mathbf{S}, \quad S = \partial V.$$

$$\iint_S \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \, \mathrm{d}S = \oint_{\partial S} P \, \mathrm{d}x + Q \, \mathrm{d}y.$$

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathrm{d}\mathbf{S} = \oint_{\ell} \mathbf{F} \cdot \mathrm{d}\boldsymbol{\ell}, \quad \ell = \partial S.$$